The Ottawa Hospital | L'Hôpital d'Ottawa

PHYSICIAN'S ORDERS
Emergency Department

Medication Allergies/Reactions	Substances or Food Allergies/Reactions
❑ None known	❑ None known

Diabetic Ketoacidosis (DKA) Protocol — Emergency

Instructions to Physician

Inclusion Criteria (need at least 2 features):
- Glucose greater than or equal to 14 mmol/L (note: glucose may be normal in DKA)
- pH less than or equal to 7.3
- Serum bicarbonate less than 22 mmol/L
- Anion gap greater than 12 mmol/L
- Positive serum beta-hydroxybutyrate or urinary ketones

Cautions:
Modifications may be required to the doses and/or rates on this form if the patient has:
- Renal failure
- Heart failure
- Age greater than 75 years

Init.	Non-Medication	Init.	IV and Medication (Medication, dose, route, frequency)

Patient Weight: _____ kg

Diabetes Type:
❑ 1 ❑ 2 ❑ Unknown

Monitoring:
☑ Resus/Emergent bed

Vital Signs:
☑ Vitals q2h **OR** other: _____
☑ Accurate ins/outs q8h

Diet:
☑ NPO

Initial Lab Investigations:
Blood:
☑ CBC, Na, K, Cl, CO_2, Mg, Ca, PO_4, Albumin, anion gap, creatinine, urea, HbA1c, CK, TNI
☑ Venous blood gas, serum osmolality, serum beta-hydroxybutyrate
❑ Blood C+S × 2

Urine:
☑ R+M, C+S
❑ beta-hCG (if female)

Diagnostics:
☑ ECG
☑ Chest x-ray

Ongoing Investigations:
☑ Blood glucose by POCT q1h
☑ Na, K, Cl, CO_2, anion gap;
Venous blood gas: ❑ q1h ❑ q2h
❑ Other: _____

☑ Establish 2 to 3 peripheral IV lines

Fluid Resuscitation (no KCl added) (caution in heart failure and renal failure):
After initial NS bolus (1–2 L), the recommended rate of NS is 500 mL/h × 4 hours followed by NS 250 mL/h × 4 hours.
❑ NS IV bolus _____ L over _____ minutes then,
❑ NS IV 500 mL/h × 4 hours then,
❑ NS IV 250 mL/h × 4 hours
❑ Maintenance fluids: _____
❑ Other: _____
☑ When blood glucose less than 14 mmol/L, start D5W at 100 mL/h (give in addition to fluid replacement if fluid resuscitation not complete)

Potassium (use separate IV line or y-site into above fluids):
Initiate potassium replacement when K less than 5.5 mmol/L.
Start after first litre of NS.

Recommendations (caution in renal failure):

K 5.5 mmol/L or greater:	No replacement
K 4.1 to 5.4 mmol/L:	NS with 20 mmol/L KCl IV
K 3.3 to 4.0 mmol/L:	NS with 40 mmol/L KCl IV
K less than 3.3 mmol/L:	KCl 10 mmol IV over 30 minutes × 2 and do not start IV insulin infusion, repeat serum potassium post-infusion of KCl

Initial Potassium Orders (choose one of the following options):
❑ KCl 10 mmol IV × _____ (each 10 mmol of KCl to be infused over a minimum of 30 minutes)
❑ NS with 20 mmol/L KCl at _____ mL/h
❑ NS with 40 mmol/L KCl at _____ mL/h
❑ Other: _____

☑ Call Physician if potassium less than 3.5 mmol/L or greater than 5 mmol/L with subsequent bloodwork
☑ Subsequent potassium replacement should be individualized according to patient response and written as a new physician's order

Analgesia:
❑ Acetaminophen 325–650 mg PO q4h prn

Antiemetics:
❑ Ondansetron 4–8 mg PO/IV q6h prn
❑ Dimenhydrinate 25–50 mg PO/IV q4h prn

Date (yy/mm/dd)	Time	Physician (printed)	Signature (Physician)
Date (noted)	Time	Processed by	Signature (Nurse)

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Init.	Diabetic Ketoacidosis (DKA) Insulin Infusion Protocol

Patient weight: _____ kg

- Start insulin regular IV infusion at _____ units/h (MD to calculate starting dose, **usual starting dose is 0.1 units/kg/h**) once potassium is greater than or equal to 3.3 mmol/L
- In ER, AMA or General Medicine Ward mix 50 units regular insulin in 250 mL NS (0.2 units/mL)
- Discontinue SC insulin
- If patient is wearing an insulin pump, discontinue pump

Insulin Infusion Adjustment:

1. Target range for blood glucose is 12–14 mmol/L.
2. Insulin infusion rate not to exceed 10 units/hour without notifying MD.
3. Measure blood glucose by POCT q1h until 2 glucose levels in target range, then q2h.
4. Resume q1h monitoring any time the insulin infusion rate is changed.
5. Document POCT blood glucose results on NUR 133.

	Blood Glucose (mmol/L)	Insulin Infusion Adjustment
	Less than 4 mmol/L	• If glucose less than 4 mmol/L give 50 mL of D5W IV over 5 minutes • Decrease insulin infusion rate by 50%, call MD and recheck blood glucose in 15 minutes • Ensure D5W IV infusion running
	4–11.9 mmol/L	• Decrease insulin infusion rate by 50% • Call MD to reassess IV fluids and insulin • Ensure D5W IV infusion running
	12–14 mmol/L (target)	• **Continue insulin infusion at current rate** • Ensure D5W IV infusion running
	14.1–16 mmol/L	• If blood glucose is less than the previous test, no rate change • If blood glucose is the same or greater than the previous test, increase the rate by 0.5 units/hour
	16.1–18 mmol/L	• If blood glucose is less than the previous test, no rate change • If blood glucose is the same or greater than the previous test, increase the rate by 1 unit/hour
	18.1–20 mmol/L	• Increase infusion rate by 2 units/hour • If blood glucose is greater than 18.1 mmol/L for 3 consecutive q1h tests, call MD to reevaluate insulin infusion rate
	Greater than 20 mmol/L	• Increase infusion rate by 3 units/hour and call MD

- **When changing patient from IV insulin to SC insulin, discontinue IV insulin 2 hours after the first dose of long-acting/premix SC insulin has been administered or 2 hours after SC insulin pump has been restarted**.
- Physicians to use SPO 215 when ready to change to SC insulin. If not on an insulin pump, a basal insulin is required for Type 1 diabetes and Type 2 diabetes on insulin at home.

Date (yy/mm/dd)	Time	Physician (printed)	Signature (Physician)
Date (noted)	Time	Processed by	Signature (Nurse)